

OC Core 1.3.3 Master Monograph

Bounded external-review scientific release

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1 Ontology of Continua Core 1.3.3

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1.1 Release Boundary

OC Core 1.3.3 is a bounded external-review scientific release. It contains a typed model foundation, theorem/proof evidence, a Lean-checked subset, finite-model semantics, target-blind numeric reconstruction rows, comparator positioning, adversarial-review closure, and journal owner-review packets.

The release does not claim final completion of every future scientific projection. It does not submit journal packages. It does not claim universal superiority over all modern science. Those broader ambitions remain in the background research program and require additional evidence before public promotion.

The public GitHub and Zenodo publication is owner-approved for this release phase. Journal submissions, email campaigns, and Software Heritage actions require separate approval.

1.2 Claim Governance

The public claim surface is intentionally bounded: a row is promoted only when it is evidence-bound, review-clean, and explicitly scoped. Universal closure or all-domain superiority language is not promoted by this release. - **claim total: 18 - unsupported promoted total: 0 - scientific promotion allowed total: 10 - absolute overclaim policy: BLOCK_PUBLIC_PROMOTION**

1.2.1 T133-K0-RES

- **claim:** K0 support is treated as a owner-review formal release-consistency check over declared resolution quotients: same-resolution states are not distinguished, and a finite countermodel shows raw separation need not induce resolution distinction. It is not promoted as an independent novelty or scientific theorem in v12.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-K0-RES.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.2 T133-OMEGA-STATUS

- **claim:** Death blocks live status; residue and rebirth are token-bound evidence relations with distinct class-specific endpoint rules: residue separates source from residue while returning to the source endpoint, and rebirth separates source, residue, and new target tokens. Rebirth is non-identity unless endpoint-bound identity evidence has identity class, declared invariant preservation, no residue token, and equal source/target endpoint evidence. No categorical Hom/composition theorem is promoted.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-OMEGA-STATUS.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.3 T133-K-ZERO

- **claim:** Continuumness zero requires live support, an independently clear obstruction ledger, and a declared zero-cause family; a zero-cause label alone does not compute $k=0$.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-K-ZERO.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.4 T133-BOUNDARY

- **claim:** Metric thresholds are a specialization of typed classifier boundaries.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-BOUNDARY.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.5 T133-HYBRID

- **claim:** OC operators are typed update semantics; chart-labelled flow-one notation is admitted only for declared chart records, while proof/rewrite and guard/reset updates remain first-class non-smooth cases. No differentiability or ODE-solution theorem is promoted in v12.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-HYBRID.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.6 T133-DIM

- **claim:** Historical axis activation and effective working rank are kept as distinct owner-review formal release-consistency fields; a finite/Lean witness shows compatibility of monotone historical bookkeeping with decreasing effective rank, but v12 does not promote an independent scientific dimension theorem.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-DIM.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.7 T133-CYCLE

- **claim:** Declared eligible-live status requires an explicit cycle mode or non-vacuous maintenance predicate.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-CYCLE.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.8 T133-ID

- **claim:** Identity continuation requires endpoint-bound identity evidence: identity class, declared invariant preservation, no residue token, equal source/target endpoint evidence, lifecycle identity-invariant truth, and typed source/target binding. Residue and rebirth evidence classes do not become identity continuation merely by preserving some invariants. No categorical Hom/composition theorem is promoted.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-ID.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.9 T133-MIN

- **claim:** Within the declared v12 release tuple semantics, each tuple component has a one-field semantic keep/drop witness that changes the release verdict.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-MIN.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.10 T133-KLEVEL

- **claim:** Every declared adjacent K-level transition $K0 \rightarrow K12$ has a release-atlas row, retained-witness evaluator check, executable finite row, and inert-witness demotion control inside the v12 release classifier; independent semantic irreducibility beyond this declared classifier is a future proof obligation, not a promoted v12 theorem.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-KLEVEL.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.2.11 OC133-NUM-PHYS-C

- **claim:** calibration replay of an official constant, not a new law of physics
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.2.12 OC133-NUM-CHEM-WEBBOOK-H2O

- **claim:** NIST Chemistry WebBook molecular-weight field replay for water only
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.2.13 OC133-NUM-CHEM-H2O

- **claim:** PubChem molecular-weight field replay for water only
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.2.14 OC133-NUM-BIO-GEO-COUNT

- **claim:** official GEO query-count replay; no organism-wide mechanism claim
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.2.15 OC133-NUM-SYS-WDI-GDP

- **claim:** retrospective WDI GDP snapshot replay QA with descriptive comparator, no prediction or superiority claim
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER

- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.2.16 OC133-NUM-MATH-FINITE

- **claim:** finite witness acceptance count for machine-checked v12 theorem inventory rows; never public-promotion, empirical, or prediction support in v12 unless a future prospective protocol explicitly flips `release_promotion_allowed`, `prediction_support_allowed`, and `empirical_support_allowed` after review
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.2.17 OC133-NOVELTY-001

- **claim:** Prior-art comparison is an illustrative bounded positioning note; uniqueness, priority, and absence are not promoted until a systematic search exists.
- **support:** ILLUSTRATIVE_PRIOR_ART_POSITIONING_ONLY_NO_UNIQUENESS_PROMOTION
- **evidence:** comparators/OC_1_3_3_NOVELTY_AND_PRIORITY_REGISTER.json
- **status:** NOT_PROMOTED_RESEARCH_NOTE_V12
- **scope limit:** No uniqueness, priority, absence, or invention claim is promoted by this row; systematic search remains future work.

1.2.18 OC133-NOSEND-001

- **claim:** Publication, DOI minting, repository release, deposit, and journal submission remain locked until both separate owner approval and an explicit manifest/channel unlock are present.
- **support:** OWNER_GATED_OWNER_REVIEW_LOCKED_CONTROL_PLANE
- **evidence:** releases/oc_core_1_3_3/editorial/OC_CORE_1_3_3_PUBLISH_MANIFEST_DRAFT.json
- **status:** GOVERNANCE_CONTROL_OWNER_REVIEW_LOCKED_NOT_SCIENTIFIC_PROMOTION
- **scope limit:** Local owner-review package only; this is not a real-world enforcement guarantee against manual or external publication. Owner approval alone is insufficient unless deposit-ready metadata, a public-record target, owner-review unlock, and every channel lock are all separately true.

1.3 Theorem and Proof Registry

The release promotes bounded theorem claims only where the theorem registry binds a proof sheet, Lean reference, finite witness route, and explicit scope boundary. Broad full-science and universal superiority claims stay outside the promoted release surface. - **registered theorem total:** 10 - **machine-checked subset total:** 10 - **adversarial blocker total:** 0

1.3.1 1. T133-K0-RES: K0 resolution-relative distinguishability theorem

- **claim boundary:** K0 support is treated as a owner-review formal release-consistency check over declared resolution quotients: same-resolution states are not distinguished, and a finite countermodel shows raw separation need not induce resolution distinction. It is not promoted as an independent novelty or scientific theorem in v12.
- **evidence ref:** appendix/OC_1_3_3_K0_RESOLUTION_FOUNDATION.tex
- **proof sheet:** proofs/proof_sheets/T133-K0-RES.md
- **Lean ref:** formal/lean/OC133V12.lean::k0_countermodel_raw_separation_not_resolution_distinction
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A proof that assumes every pair of raw real states is epsilon-separated is outside v12 and fails G33.

Proof-sheet excerpt:

2 T133-K0-RES - K0 resolution-relative distinguishability theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: appendix/OC_1_3_3_K0_RESOLUTION_MACHINE_CHECKED_SUBSET Machine-checked subset: formal/lean/OC133V12.lean::k0_countermodel_raw_separation_not_resolution_distinction Attacked claim: K0 support is treated as a owner-review formal release-consistency check over declared resolution quotients: same-resolution states are not distinguished, and a finite countermodel shows raw separation need not induce resolution distinction. It is not promoted as an independent novelty or scientific theorem in v12. ## Assumptions - A raw carrier may be continuous, finite, countable, graph-like, proof-theoretic, or typed-combinatorial. - A resolution regime supplies an observational equivalence relation over raw states. - Separation is asserted only for resolved quotient classes and only inside the declared regime. ## Definitions - Resolved carrier S/ρ : raw states modulo the observational equivalence induced by ρ . - Distinguishable state: two quotient classes are unequal under the declared resolution. - K0 support: bookkeeping distinguishability for a declared model state, not ontology of raw atoms. ## Lemma 1 If two raw points are in the same ρ -cell, no OC theorem may infer raw separation between them. ## Lemma 2 If two ρ -cells are distinct and the quotient metric declares positive separation, K0 distinguishability follows without a raw lower bound. ## Theorem K0 is compatible with continuous raw carriers inside the declared release semantics because the required separation is a quotient property. ## Proof The proof factors every K0 reference through ρ . Lemma 1 blocks raw discreteness leakage. Lemma 2 supplies the only positive separation used by downstream K0 checks. Therefore the owner-review formal check is about resolved classes, not raw points. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary A proof that assumes every pair of raw real states is epsilon-separated is outside v12 and fails G33. ## Machine-Checkable Finite Example Partition $[0,1]$ into four cells. Points 0.10 and 0.11 remain unresolved, while the first and second cells are separated as quotient classes. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - appendix/OC_1_3_3_K0_RESOLUTION_FOUNDATION.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

2.0.1 2. T133-OMEGA-STATUS: Typed liveness, death, residue, and rebirth evidence-consistency theorem

- **claim boundary:** Death blocks live status; residue and rebirth are token-bound evidence relations with distinct class-specific endpoint rules: residue separates source from residue while returning to the source endpoint, and rebirth separates source, residue, and new target tokens. Rebirth is non-identity unless endpoint-bound identity evidence has identity class, declared invariant preservation, no residue token, and equal source/target endpoint evidence. No categorical Hom/composition theorem is promoted.
- **evidence ref:** content/OC_1_3_3_TYPED_FOUNDATION.tex
- **proof sheet:** proofs/proof_sheets/T133-OMEGA-STATUS.md
- **Lean ref:** formal/lean/OC133V12.lean::lifecycle_residue_rebirth_morphism_boundary
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** Any claim reading residue-preserving restart as same-identity survival is rejected unless endpoint-bound identity evidence is supplied.

Proof-sheet excerpt:

3 T133-OMEGA-STATUS - Typed liveness, death, residue, and rebirth evidence-consistency theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: content/OC_1_3_3_TYPED_FOUNDATION Machine-checked subset: formal/lean/OC133V12.lean::lifecycle_residue_rebirth_morphism_boundary
Attacked claim: Death blocks live status; residue and rebirth are token-bound evidence relations with distinct class-specific endpoint rules: residue separates source from residue while returning to the source endpoint, and rebirth separates source, residue, and new target tokens. Rebirth is non-identity unless endpoint-bound identity evidence has identity class, declared invariant preservation, no residue token, and equal source/target endpoint evidence. No categorical Hom/composition theorem is promoted. ## Assumptions - Admissibility, liveness, death, residue, rebirth, and identity are separate typed fields. - Every realization declares the predicates that can change live status. - Residue preservation is not identity continuation unless endpoint-bound identity evidence is present: identity class, preserved invariants, no residue token, and equal source/target endpoints. ## Definitions - Live(K,t): typed boolean status over a realization. - Residue(K,t): preserved post-death structure with its own carrier. - Rebirth morphism: a construction from residue into a new live realization. ## Lemma 1 Nonempty admissibility does not imply liveness without the live-support predicates. ## Lemma 2 Residue preservation does not imply identity continuation without identity morphism constraints. ## Theorem The four statuses are jointly consistent and non-interchangeable in the typed OC model, and the residue/rebirth source-target route is bound to explicit token evidence. ## Proof The fields have distinct codomains and transition rules. Lemma 1 separates admissibility from liveness. Lemma 2 separates residue from identity. The Lean theorem then takes explicit source, residue, and target tokens: residue evidence separates source from residue while returning to the source endpoint, and rebirth evidence carries pairwise source/residue/new-target separation. This proves death blocks liveness while residue/rebirth evidence is not endpoint-bound identity evidence. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary Any claim reading residue-preserving restart as same-identity survival is rejected unless endpoint-bound identity evidence is supplied. ## Machine-Checkable Finite Example A two-state automaton has admissible state A, failed cycle support, residue r, and new state B constructed from r; B is rebirth, not continuation. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - content/OC_1_3_3_TYPED_FOUNDATION.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

3.0.1 3. T133-K-ZERO: Continuumness zero obstruction theorem

- **claim boundary:** Continuumness zero requires live support, an independently clear obstruction ledger, and a declared zero-cause family; a zero-cause label alone does not compute $k=0$.
- **evidence ref:** appendix/OC_1_3_3_CONTINUUMNESS_FUNCTIONALS.tex
- **proof sheet:** proofs/proof_sheets/T133-K-ZERO.md
- **Lean ref:** formal/lean/OC133V12.lean::continuumness_zero_case_iff_declared_zero_cause_with_support
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A realization with undeclared zero-cause, missing live support, or any active obstruction may not promote $k=0$.

Proof-sheet excerpt:

4 T133-K-ZERO - Continuumness zero obstruction theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: appendix/OC_1_3_3_CONTINUUMNESS_MACHINE_CHECKED_SUBSET Machine-checked subset: formal/lean/OC133V12.lean::continuumness_zero_case_iff_declared_zero_cause_with_claim Attacked claim: Continuumness zero requires live support, an independently clear obstruction ledger, and a declared zero-cause family; a zero-cause label alone does not compute $k=0$. ## Assumptions - The continuumness score k is separate from live status. - Zero-cause predicates are explicitly declared for each realization. - A product formula is a local representation only when independence assumptions are stated. ## Definitions - ZeroCause(K,t): disjunction of typed collapse causes. - ObstructionLedger(K,t): typed flow/coherence/identity/embedding obstruction flags. - $k(K,t)=0$: score-zero event computed from the obstruction ledger and licensed by at least one active zero-cause. - Local aggregator: product or other numeric representation derived after semantics are fixed. ## Lemma 1 Flow-support collapse can make k zero while admissibility and cycles are nonempty. ## Lemma 2 Coherence contradiction can make k zero without set emptiness. ## Theorem Within v12, $k=0$ is equivalent to nonempty support, an active declared zero-cause, and no active obstruction in the independent obstruction ledger. ## Proof The score is computed from the obstruction ledger, not from the zero-cause label. Lemma 1 proves zero score iff no obstruction is active. Lemma 2 proves a zero-cause with clear obstruction licenses the zero verdict, while an active obstruction rejects $k=0$ even if a zero-cause label exists. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary A realization with undeclared zero-cause, missing live support, or any active obstruction may not promote $k=0$. ## Machine-Checkable Finite Example $\Omega=\{s\}$, $C=\{c\}$, flow_zero_cause=true, and all obstruction flags false gives $k=0$; a matching obstruction-active control rejects $k=0$ despite the zero-cause label. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - appendix/OC_1_3_3_CONTINUUMNESS_FUNCTIONALS.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

4.0.1 4. T133-BOUNDARY: Metric-threshold boundary specialization theorem

- **claim boundary:** Metric thresholds are a specialization of typed classifier boundaries.
- **evidence ref:** appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex
- **proof sheet:** proofs/proof_sheets/T133-BOUNDARY.md
- **Lean ref:** formal/lean/OC133V12.lean::metric_boundary_specialization
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A social or logical boundary represented numerically without a measurement rule is blocked by G40.

Proof-sheet excerpt:

5 T133-BOUNDARY - Metric-threshold boundary specialization theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_MACHINE_CHECKED_SUBSET Machine-checked subset: formal/lean/OC133V12.lean::metric_boundary_specialization Attacked claim: Metric thresholds are a specialization of typed classifier boundaries. ## Assumptions - Boundary predicates are typed classifiers into status objects. - Metric thresholds are admitted only when the domain supplies a metric measurement rule. - Logical, categorical, graph, social, and proof-state boundaries may remain non-metric. ## Definitions - Classifier boundary: $b_i:S \rightarrow \text{Status}_i$ plus a failure predicate over Status_i . - Metric boundary: classifier with $\text{Status}_i = \text{real-valued}$ or

ordered numeric status. - Logical boundary: classifier with `Status_i = {true,false}`. `## Lemma 1` Every real-valued threshold boundary embeds as a classifier boundary. `## Lemma 2` A boolean admissibility rule is a classifier boundary without inventing a fake numeric distance. `## Theorem` The v12 boundary formalism conservatively extends metric-threshold OC boundaries. `## Proof` Map each threshold measurement to a classifier returning its measured status and use the threshold comparison as the failure predicate. Non-metric domains instantiate the same classifier type directly. Thus old metric cases are preserved and non-metric cases stop pretending to be metric. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. `## Counterexample Boundary` A social or logical boundary represented numerically without a measurement rule is blocked by G40. `## Machine-Checkable Finite Example` A proof state is admissible iff `Consistent(state)=true`; no real-valued boundary is required. `## Dependency` Refs - `proofs/THEOREM_INVENTORY_1_3_3.json` - `proofs/FINITE_MODEL_CHECKS_1_3_3.json` - `formal/lean/OC133V12.lean` - `appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex` `## Reviewer Attack Answered` The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

5.0.1 5. T133-HYBRID: Typed update and chart-labelled operator semantics theorem

- **claim boundary:** OC operators are typed update semantics; chart-labelled flow-one notation is admitted only for declared chart records, while proof/rewrite and guard/reset updates remain first-class non-smooth cases. No differentiability or ODE-solution theorem is promoted in v12.
- **evidence ref:** `content/OC_1_3_3_OPERATOR_SEMANTICS.tex`
- **proof sheet:** `proofs/proof_sheets/T133-HYBRID.md`
- **Lean ref:** `formal/lean/OC133V12.lean::integrated_operator_semantics`
- **evidence ceiling:** `BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS`
- **scope limit:** Any route treating a chart token as differentiability, manifold, vector-field, or ODE-solution evidence is outside the v12 operator theorem and remains a future lint/proof obligation.

Proof-sheet excerpt:

6 T133-HYBRID - Typed update and chart-labelled operator semantics theorem

Status: `PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED` Primary artifact: `content/OC_1_3_3_OPERATOR_SEMANTICS.tex` Machine-checked subset: `formal/lean/OC133V12.lean::integrated_operator_semantics` Attacked claim: OC operators are typed update semantics; chart-labelled flow-one notation is admitted only for declared chart records, while proof/rewrite and guard/reset updates remain first-class non-smooth cases. No differentiability or ODE-solution theorem is promoted in v12. `## Assumptions` - Operators are typed update components over realization states. - A derivative claim is not licensed by this v12 theorem; chart records only gate flow-one notation in the typed update subset. - The promoted formal subset covers smooth-chart updates, proof/rewrite updates, and guard/reset hybrid updates; stochastic and graph operators remain unpromoted extension obligations until separately formalized. `## Definitions` - Update semantics: state and admissible input map to a successor object or distribution. - Chart-labelled semantics: update may carry chart metadata, but differentiability is not promoted without a separate theorem. - Hybrid semantics: smooth segments and discrete jumps live in one typed transition system. `## Lemma 1` A declared chart-labelled flow-one route induces a typed update relation in the v12 subset. `## Lemma 2` A typed update relation need not induce a derivative without extra smoothness assumptions. `## Theorem` OC operators F,G,H,Q,R,S,U are typed updates; chart-labelled flow-one, proof/rewrite, and guard/reset hybrid routes are separate typed realizations. `## Proof` The primitive object is a route-specific operator-

admission record. The Lean theorem `integrated_operator_semantics` binds the smooth-chart flow-one update, chart/domain/local-law obligations, guard/reset update branch, and proof/rewrite non-smooth branch in one statement; `operator_admission_route_obligations` remains a supporting decomposition lemma. The Lean theorem `bound_hybrid_admission_obligations` adds a system/source-bound hybrid admission record, tying guard observation, guard value, reset source/target typing, codomain typing, and reset admissibility to the actual `HybridSystem` and source state. The false-guard pass-through branch is checked by `hybrid_no_guard_uses_update` and by finite rows whose operator admission records guard evaluation rather than assuming the guard value is true. The finite runner independently evaluates the same fields and fails if labels are correct but obligations are missing. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. `## Counterexample Boundary` Any route treating a chart token as differentiability, manifold, vector-field, or ODE-solution evidence is outside the v12 operator theorem and remains a future lint/proof obligation. `## Machine-Checkable Finite Example` The finite corpus includes smooth-chart positive/negative cases, guard/reset positive/negative cases, proof/rewrite positive/negative cases, and tamper controls for missing local law, wrong reset codomain, and derivative leakage. `## Dependency Refs` - `proofs/THEOREM_INVENTORY_1_3_3.json` - `proofs/FINITE_MODEL_CHECKS_1_3_3.json` - `formal/lean/OC133V12.lean` - `content/OC_1_3_3_OPERATOR_SEMANTICS.tex` `## Reviewer Attack Answered` The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

6.0.1 6. T133-DIM: Historical axis and effective-rank compatibility theorem

- **claim boundary:** Historical axis activation and effective working rank are kept as distinct owner-review formal release-consistency fields; a finite/Lean witness shows compatibility of monotone historical bookkeeping with decreasing effective rank, but v12 does not promote an independent scientific dimension theorem.
- **evidence ref:** `appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex`
- **proof sheet:** `proofs/proof_sheets/T133-DIM.md`
- **Lean ref:** `formal/lean/OC133V12.lean::historical_axis_survives_rank_drop`
- **evidence ceiling:** `BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS`
- **scope limit:** A K-level is demotable only when the alleged new axis has no witness and no observable consequence.

Proof-sheet excerpt:

7 T133-DIM - Historical axis and effective-rank compatibility theorem

Status: `PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED` Primary artifact: `appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex`
Machine-checked subset: `formal/lean/OC133V12.lean::historical_axis_survives_rank_drop`
Attacked claim: Historical axis activation and effective working rank are kept as distinct owner-review formal release-consistency fields; a finite/Lean witness shows compatibility of monotone historical bookkeeping with decreasing effective rank, but v12 does not promote an independent scientific dimension theorem. `## Assumptions` - Historical axes record realized dependence history. - Effective rank records currently active independent degrees of freedom. - A lawful demotion must prove the historical axis is unobservable under the declared equivalence. `## Definitions` - `A_hist`: set of historically activated axes. - `rank_eff(t)`: rank of currently active support. - Axis witness: a finite pair whose verdict changes when the axis is removed. `## Lemma 1` A frozen memory axis can remain historically present after active rank falls. `## Lemma 2` Reduction from

K(n+1) to K(n) fails when a witness remains observable. `## Theorem Historical monotonicity and effective-rank decrease are compatible inside the release semantics because they measure different typed quantities. ## Proof A_hist is accumulated over realized dependence events; rank_eff is recomputed over active support. Lemma 1 gives compatibility; Lemma 2 gives the irreducibility test used by the atlas. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary A K-level is demotable only when the alleged new axis has no witness and no observable consequence. ## Machine-Checkable Finite Example A two-axis automaton activates memory and later freezes it; historical axes remain two while active rank becomes one. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.`

7.0.1 7. T133-CYCLE: Live-status cycle-mode requirement theorem

- **claim boundary:** Declared eligible-live status requires an explicit cycle mode or non-vacuous maintenance predicate.
- **evidence ref:** `content/OC_1_3_3_CYCLE_TAXONOMY.tex`
- **proof sheet:** `proofs/proof_sheets/T133-CYCLE.md`
- **Lean ref:** `formal/lean/OC133V12.lean::eligible_live_requires_cycle_or_maintenance`
- **evidence ceiling:** `BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS`
- **scope limit:** An artifact that never updates, replays, checks, or maintains itself is archive residue, not live continuum.

Proof-sheet excerpt:

8 T133-CYCLE - Live-status cycle-mode requirement theorem

Status: `PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED` Primary artifact: `content/OC_1_3_3_CYCLE_TAXONOMY`
Machine-checked subset: `formal/lean/OC133V12.lean::eligible_live_requires_cycle_or_maintenance`
Attacked claim: Declared eligible-live status requires an explicit cycle mode or non-vacuous maintenance predicate. `## Assumptions - Live status is not static persistence. - Maintenance, renewal, replay, regulatory, and degenerate cycle modes are distinct. - A degenerate fixed point is live only if its maintenance predicate is non-vacuous. ## Definitions - Cycle mode: typed recurrence, maintenance, replay, or regulation condition. - Frozen persistence: status that stays unchanged without support obligation. - Degenerate maintenance: identity recurrence plus active support checks. ## Lemma 1 No declared cycle mode means the live predicate is under-specified. ## Lemma 2 A fixed point with active support obligations can satisfy degenerate maintenance. ## Theorem OC live status requires explicit cycle evidence; static labels are residue or inert records. ## Proof Liveness is defined through support that can fail or be maintained. Lemma 1 rejects unsupported static labels. Lemma 2 admits legitimate fixed points. The theorem follows by the typed live predicate. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary An artifact that never updates, replays, checks, or maintains itself is archive residue, not live continuum. ## Machine-Checkable Finite Example A constant automaton with an energy-maintenance check passes; a label with no check fails. ## Dependency`

Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - content/OC_1_3_3_CYCLE_TAXONOMY.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

8.0.1 8. T133-ID: Identity, residue, and rebirth evidence-classification theorem

- **claim boundary:** Identity continuation requires endpoint-bound identity evidence: identity class, declared invariant preservation, no residue token, equal source/target endpoint evidence, lifecycle identity-invariant truth, and typed source/target binding. Residue and rebirth evidence classes do not become identity continuation merely by preserving some invariants. No categorical Hom/composition theorem is promoted.
- **evidence ref:** appendix/OC_1_3_3_IDENTITY_RESIDUE_REBIRTH_CLASSIFICATION.tex
- **proof sheet:** proofs/proof_sheets/T133-ID.md
- **Lean ref:** formal/lean/OC133V12.lean::endpoint_bound_identity_classification
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** Any public claim reading rebirth as literal same-identity survival is blocked.

Proof-sheet excerpt:

9 T133-ID - Identity, residue, and rebirth evidence-classification theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: appendix/OC_1_3_3_IDENTITY_RES Machine-checked subset: formal/lean/OC133V12.lean::endpoint_bound_identity_classification
Attacked claim: Identity continuation requires endpoint-bound identity evidence: identity class, declared invariant preservation, no residue token, equal source/target endpoint evidence, lifecycle identity-invariant truth, and typed source/target binding. Residue and rebirth evidence classes do not become identity continuation merely by preserving some invariants. No categorical Hom/composition theorem is promoted. ## Assumptions - Identity continuation, residue preservation, and rebirth are separate evidence classes with explicit source/target endpoint evidence. - The realization declares invariants identity morphisms must preserve. - No metaphysical personal-identity claim is promoted by the formal morphism class alone. ## Definitions - Endpoint-bound identity evidence: preserves declared identity invariants, has no residue token, and binds equal source/target endpoints. - Residue evidence: preserves a proper subset sufficient for reconstruction evidence. - Rebirth evidence: maps residue into a new live realization. ## Lemma 1 Residue morphisms compose with rebirth constructors without becoming identity morphisms. ## Lemma 2 If an identity invariant is absent after restart, the morphism class is residue or rebirth, not identity. ## Theorem The v12 morphism classes block residue/rebirth identity equivocation. ## Proof The classifier has an explicit positive endpoint-bound identity case and negative residue/rebirth cases. Invariant preservation alone is insufficient for residue or rebirth classes, identity class alone is insufficient without preserved invariants, and preserved invariants are still insufficient when source/target endpoints mismatch or a residue token is present. The finite runner exhausts morphism class x invariant-preserved x endpoint-equality x residue-token x claimed-identity truth-table rows. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary Any public claim reading rebirth as literal same-identity survival is blocked. ## Machine-Checkable Finite Example A process checkpoint preserves schema and loses runtime token identity; restart is rebirth, not same identity. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - appendix/OC_1_3_3_IDENTITY_RESIDUE_REBIRTH_CLASSIFICATION.tex

Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

9.0.1 9. T133-MIN: Declared semantic-verdict component independence theorem

- **claim boundary:** Within the declared v12 release tuple semantics, each tuple component has a one-field semantic keep/drop witness that changes the release verdict.
- **evidence ref:** appendix/OC_1_3_3_GLOBAL_MINIMALITY_WITNESSES.tex
- **proof sheet:** proofs/proof_sheets/T133-MIN.md
- **Lean ref:** formal/lean/OC133V12.lean::release_tuple_semantic_component_irredundant
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** If any declared tuple component can be removed while FM-MIN-* still returns PASS for the declared semantic verdict suite, the minimality card fails.

Proof-sheet excerpt:

10 T133-MIN - Declared semantic-verdict component independence theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: appendix/OC_1_3_3_GLOBAL_MINIMALITY_WITNESSES.tex

Machine-checked subset: formal/lean/OC133V12.lean::release_tuple_semantic_component_irredundant

Attacked claim: Within the declared v12 release tuple semantics, each tuple component has a one-field semantic keep/drop witness that changes the release verdict. ## Assumptions - Minimality is claimed for the release-governed OC verdict class, not for all possible theories. - Each declared component has a witness pair that changes a declared OC verdict when the component is removed or weakened. - Witnesses are checked by the finite-model ledger and by the Lean component-witness schema. ## Definitions - Verdict-invariant: preserves pass/fail classification of the declared OC tests. - Witness pair: two cases differing only in one component and producing different verdicts. - Global tuple minimality: every declared tuple component has at least one witness pair. ## Lemma 1 A component with a verdict-changing witness cannot be removed verdict-invariantly. ## Lemma 2 The v12 witness ledger covers every declared tuple component. ## Theorem The declared v12 tuple has component-wise independence for the declared semantic verdict suite. ## Proof For each component c, the witness ledger gives keep_c and drop_c cases whose semantic records differ only in c's obligation field and whose verdicts differ. Lemma 1 proves that c is required for the release verdict suite. Lemma 2 ranges over the full declared tuple. Therefore no declared component can be removed while preserving this declared v12 verdict suite. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary If any declared tuple component can be removed while FM-MIN-* still returns PASS for the declared semantic verdict suite, the minimality card fails. ## Machine-Checkable Finite Example Removing boundary admits a state rejected by the full tuple; removing cycle mode admits a frozen non-live object. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - appendix/OC_1_3_3_GLOBAL_MINIMALITY_WITNESSES.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

10.0.1 10. T133-KLEVEL: Declared adjacent K-level atlas/evaluator consistency theorem

- **claim boundary:** Every declared adjacent K-level transition K0->K12 has a release-atlas row, retained-witness evaluator check, executable finite row, and inert-witness demotion control inside the v12 release

classifier; independent semantic irreducibility beyond this declared classifier is a future proof obligation, not a promoted v12 theorem.

- **evidence ref:** appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex
- **proof sheet:** proofs/proof_sheets/T133-KLEVEL.md
- **Lean ref:** formal/lean/OC133V12.lean::release_atlas_manifest_has_total_finite_case_coverage
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** If any adjacent K row lacks row identity, retained-witness evaluator failure, or inert-witness demotion control, the finite negative control fails; independent semantic irreducibility remains unpromoted unless separately proven.

Proof-sheet excerpt:

11 T133-KLEVEL - Declared adjacent K-level atlas/evaluator consistency theorem

Status: PROMOTED_BOUNDED_THEOREM_V12_OWNER_REVIEW_LOCKED Primary artifact: appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex
Machine-checked subset: formal/lean/OC133V12.lean::release_atlas_manifest_has_total_finite_case_coverage
Attacked claim: Every declared adjacent K-level transition $K0 \rightarrow K12$ has a release-atlas row, retained-witness evaluator check, executable finite row, and inert-witness demotion control inside the v12 release classifier; independent semantic irreducibility beyond this declared classifier is a future proof obligation, not a promoted v12 theorem. ## Assumptions - K-levels are release-governed classifier levels, not metaphysical ranks. - Adjacent row consistency is asserted only inside the declared v12 release classifier. - A lawful demotion is allowed when the witness disappears under a stronger equivalence or becomes observationally inert. ## Definitions - Adjacent transition witness: finite pair that flips verdict when the added K-axis is removed. - Reduction-failure criterion: declared evaluator condition under which $K(n+1)$ cannot be represented at $K(n)$ inside the release classifier. - Lawful demotion criterion: condition under which $K(n+1)$ may be treated as $K(n)$ without verdict loss. ## Lemma 1 A transition with an observable witness cannot be reduced without verdict loss. ## Lemma 2 A transition with no observable witness is demotable by the stated criterion rather than inflated. ## Theorem The declared $K0$ - $K12$ atlas blocks reduction exactly for retained adjacent witnesses and allows demotion exactly for inert witnesses inside the v12 release classifier. ## Proof Each row in the atlas records the new axis, witness pair, reduction-failure criterion, and demotion criterion. The finite runner verifies exact row identity, adjacency, criterion text, retained witness verdict loss, demotion verdict preservation, and unique $K0 \rightarrow K12$ coverage. Lemma 1 handles retained declared witnesses. Lemma 2 handles non-retained witnesses without inflation. The atlas has zero unresolved adjacent rows inside the declared classifier; no domain-independent irreducibility theorem is promoted by this proof sheet. The proof is promoted only as a bounded owner-review release claim with the stated assumptions. It is linked to the finite witness corpus and to the Lean subset named above. The Lean item checks the corresponding typed invariant, while the finite corpus checks the release verdict behavior used by the public claim ledger. ## Counterexample Boundary If any adjacent K row lacks row identity, retained-witness evaluator failure, or inert-witness demotion control, the finite negative control fails; independent semantic irreducibility remains unpromoted unless separately proven. ## Machine-Checkable Finite Example $K3$ autocatalytic closure cannot be represented by $K2$ phase threshold alone when closure production is the verdict-changing axis. ## Dependency Refs - proofs/THEOREM_INVENTORY_1_3_3.json - proofs/FINITE_MODEL_CHECKS_1_3_3.json - formal/lean/OC133V12.lean - appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex ## Reviewer Attack Answered The hostile attack is answered by separating type assumptions, exact theorem scope, executable witnesses, and a falsifier boundary. If a reviewer removes the assumptions, the claim is not silently weakened; the relevant v12 gate fails.

11.1 Lean Formalization Subset

The Lean subset is a machine-checked subset of the OC 1.3.3 theorem surface. The release does not claim that every mathematical or empirical statement is fully formalized in Lean. - **certificate state:** - **theorem ref total:** 10 - **missing theorem ref total:** - **Lean file:** formal/lean/OC133V12.lean

Selected declaration excerpt:

```
namespace OC133V12

inductive Status where
| pass
| fail
deriving DecidableEq, Repr

inductive CycleMode where
| maintenance
| renewal
| replay
| regulatory
| degenerate
deriving DecidableEq, Repr

inductive MorphismClass where
| identity
| residue
| rebirth
deriving DecidableEq, Repr

inductive Component where
| carrier
| realization
| lawfulPossibility
| liveness
| residue
| morphisms
| boundaries
| operators
| cycles
| dimension
| kFunctional
deriving DecidableEq, Repr

inductive AdjacentK where
| k0_k1
| k1_k2
| k2_k3
| k3_k4
| k4_k5
| k5_k6
| k6_k7
| k7_k8
| k8_k9
| k9_k10
| k10_k11
| k11_k12
deriving DecidableEq, Repr
```

```

def boolStatus (b : Bool) : Status :=
  if b then Status.pass else Status.fail

structure Resolution (S : Type) where
  cell : S -> Nat

structure RawSeparation (S : Type) where
  separated : S -> S -> Bool

def sameCell {S : Type} (rho : Resolution S) (a b : S) : Prop :=
  rho.cell a = rho.cell b

def distinguished {S : Type} (rho : Resolution S) (a b : S) : Prop :=
  rho.cell a != rho.cell b

theorem k0_same_cell_not_distinguished {S : Type} (rho : Resolution S) (a b : S) :
  sameCell rho a b -> distinguished rho a b = False := by
  intro h
  unfold distinguished
  rw [h]
  simp

theorem k0_distinguished_requires_resolved_delta {S : Type} (rho : Resolution S) (a b : S) :
  distinguished rho a b -> sameCell rho a b -> False := by
  intro hd hs
  rw [k0_same_cell_not_distinguished rho a b hs] at hd
  exact hd

theorem k0_resolution_does_not_force_raw_separation {S : Type}
  (rho : Resolution S) (raw : RawSeparation S) (a b : S) :
  sameCell rho a b -> raw.separated a b = true -> distinguished rho a b = False := by
  intro hs _
  exact k0_same_cell_not_distinguished rho a b hs

inductive K0CounterPoint where
  | a
  | b
deriving DecidableEq, Repr

def k0CounterResolution : Resolution K0CounterPoint :=
  { cell := fun _ => 0 }

def k0CounterRawSeparation : RawSeparation K0CounterPoint :=
  { separated := fun x y => decide (x != y) }

theorem k0_countermodel_raw_separation_not_resolution_distinction :
  sameCell k0CounterResolution K0CounterPoint.a K0CounterPoint.b /\
  k0CounterRawSeparation.separated K0CounterPoint.a K0CounterPoint.b = true /\
  distinguished k0CounterResolution K0CounterPoint.a K0CounterPoint.b = False := by
  exact And.intro rfl (And.intro (by decide) (k0_same_cell_not_distinguished k0CounterResolution K0CounterPoint.a K0CounterPoint.b))

structure MaintenanceEvidence where
  obligationChecked : Bool
  supportAvailable : Bool

```

```

def nonVacuousMaintenance (m : MaintenanceEvidence) : Prop :=
  m.obligationChecked = true /\ m.supportAvailable = true

structure Realization where
  Carrier : Type
  admissible : Carrier -> Bool
  live : Carrier -> Bool
  cycle : Carrier -> Option CycleMode
  maintenance : Carrier -> Option MaintenanceEvidence

def maintenanceWitnessed (R : Realization) (x : R.Carrier) : Prop :=
  exists m, R.maintenance x = some m /\ nonVacuousMaintenance m

def cycleWitnessed (R : Realization) (x : R.Carrier) : Prop :=
  match R.cycle x with
  | none => False
  | some CycleMode.degenerate => maintenanceWitnessed R x
  | some _ => True

def supportWitnessed (R : Realization) (x : R.Carrier) : Prop :=
  cycleWitnessed R x /\ maintenanceWitnessed R x

def eligibleLive (R : Realization) (x : R.Carrier) : Prop :=
  R.admissible x = true /\ R.live x = true /\ supportWitnessed R x

theorem eligible_live_requires_cycle (R : Realization) (x : R.Carrier) :
  eligibleLive R x -> supportWitnessed R x := by
  intro h
  exact h.right.right

theorem eligible_live_requires_cycle_or_maintenance (R : Realization) (x : R.Carrier) :
  eligibleLive R x -> cycleWitnessed R x /\ maintenanceWitnessed R x := by
  intro h
  exact h.right.right

theorem eligible_live_requires_admissible (R : Realization) (x : R.Carrier) :
  eligibleLive R x -> R.admissible x = true := by
  intro h
  exact h.left

theorem cycle_mode_required_for_eligible_live (R : Realization) (x : R.Carrier) :
  eligibleLive R x -> (R.maintenance x = none) -> R.cycle x != none := by
  intro h hm
  cases h.right.right with
  | inl hc =>
    unfold cycleWitnessed at hc
    cases hcyc : R.cycle x with
    | none =>
      rw [hcyc] at hc
      cases hc
      | some c =>
        simp
        | inr hmnt =>
          rcases hmnt with m, hsome, _

```

```

rw [hm] at hsome
cases hsome

structure Lifecycle (S Residue NewLive : Type) where
  admissible : S -> Bool
  live : S -> Bool
  death : S -> Bool
  deathBlocksLive : forall x : S, death x = true -> live x = false
  residueOf : S -> Option Residue
  rebirthOf : Residue -> Option NewLive
  identityInvariant : S -> NewLive -> Bool
  sourceToken : S -> Nat
  residueToken : Residue -> Nat
  newLiveToken : NewLive -> Nat
  sourceTokenInjective : forall x y : S, sourceToken x = sourceToken y -> x = y
  residueTokenInjective : forall x y : Residue, residueToken x = residueToken y -> x = y
  newLiveTokenInjective : forall x y : NewLive, newLiveToken x = newLiveToken y -> x = y

theorem declared_death_blocks_live {S Residue NewLive : Type}
  (L : Lifecycle S Residue NewLive) (x : S) :
  L.death x = true -> L.live x = false := by
  intro h
  exact L.deathBlocksLive x h

theorem death_live_conflict_impossible {S Residue NewLive : Type}
  (L : Lifecycle S Residue NewLive) (x : S) :
  L.death x = true -> L.live x = true -> False := by
  intro hdeath hlive
  have hblocked : L.live x = false := L.deathBlocksLive x hdeath
  rw [hlive] at hblocked
  cases hblocked

theorem residue_rebirth_are_typed_source_target_relations {S Residue NewLive : Type}
  (L : Lifecycle S Residue NewLive) (x : S) (r : Residue) (y : NewLive) :
  L.residueOf x = some r -> L.rebirthOf r = some y -> L.death x = true ->
  L.live x = false /\ L.residueOf x = some r /\ L.rebirthOf r = some y := by
  intro hres hreb hdeath
  exact And.intro (declared_death_blocks_live L x hdeath) (And.intro hres hreb)

theorem lifecycle_source_tokens_bind_carrier_endpoint {S Residue NewLive : Type}
  (L : Lifecycle S Residue NewLive) (x y : S) :
  L.sourceToken x = L.sourceToken y -> x = y := by
  exact L.sourceTokenInjective x y

theorem lifecycle_rebirth_tokens_bind_target_endpoint {S Residue NewLive : Type}
  (L : Lifecycle S Residue NewLive) (x y : NewLive) :
  L.newLiveToken x = L.newLiveToken y -> x = y := by
  exact L.newLiveTokenInjective x y

structure MorphismEvidence where
  sourceToken : Nat
  residueToken : Option Nat
  targetToken : Nat
  mclass : MorphismClass
  invariantPreserved : Bool

```

```

def isIdentityMorphism (m : MorphismEvidence) : Prop :=
  m.mclass = MorphismClass.identity /\
  m.invariantPreserved = true /\
  m.residueToken = none /\
  m.sourceToken = m.targetToken

```

11.2 Executable Finite-Model Evidence

The finite-model runner computes semantic verdicts from model facts and mutation controls. The public summary below omits internal publication-control rows and preserves the semantic proof evidence. - **verdict:** - **failure total:** 0 - **semantic evaluator:** True - **mutation control total:** 6 - **K-transition negative total:** 12

11.2.1 FM-T133-K0-RES-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.2 FM-T133-K0-RES-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.3 FM-T133-OMEGA-STATUS-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.4 FM-T133-OMEGA-STATUS-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.5 FM-T133-OMEGA-IDENTITY-EQUIVOCATION-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.6 FM-T133-K-ZERO-POS

- **case type:** theorem_case

- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.7 FM-T133-K-ZERO-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.8 FM-T133-K-ZERO-OBSTRUCTION-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.9 FM-T133-K-ZERO-LIVE-SUPPORT-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.10 FM-T133-BOUNDARY-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.11 FM-T133-BOUNDARY-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.12 FM-T133-HYBRID-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.13 FM-T133-HYBRID-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT

- expected verdict: REJECT
- failure total:
- witness:

11.2.14 FM-T133-HYBRID-NO-GUARD-STEP-POS

- case type: theorem_case
- observed verdict: ACCEPT
- expected verdict: ACCEPT
- failure total:
- witness:

11.2.15 FM-T133-HYBRID-NO-GUARD-STEP-NEG

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.16 FM-T133-HYBRID-GUARD-MISSING-NEG

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.17 FM-T133-HYBRID-GUARD-NONBOOLEAN-NEG

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.18 FM-T133-HYBRID-GUARD-VALUE-MISMATCH-NEG

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.19 FM-T133-HYBRID-RESET-SOURCE-MODE-NEG

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.20 FM-T133-HYBRID-RESET-TARGET-MODE-NEG

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT

- **failure total:**
- **witness:**

11.2.21 FM-T133-HYBRID-SMOOTH-CHART-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.22 FM-T133-HYBRID-SMOOTH-CHART-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.23 FM-T133-HYBRID-PROOF-UPDATE-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.24 FM-T133-HYBRID-PROOF-UPDATE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.25 FM-T133-HYBRID-SMOOTH-LOCAL-LAW-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.26 FM-T133-HYBRID-PROOF-NO-RULE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.27 FM-T133-DIM-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.28 FM-T133-DIM-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.29 FM-T133-CYCLE-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.30 FM-T133-CYCLE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.31 FM-T133-ID-REBIRTH-NONIDENTITY-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.32 FM-T133-ID-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.33 FM-T133-ID-RESIDUE-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.34 FM-T133-ID-RESIDUE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.35 FM-T133-ID-IDENTITY-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.36 FM-T133-MIN-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.37 FM-T133-MIN-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.38 FM-T133-KLEVEL-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.39 FM-T133-KLEVEL-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.40 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.41 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.42 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.43 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.44 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: ACCEPT
- expected verdict: ACCEPT
- failure total:
- witness:

11.2.45 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.46 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.47 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.48 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.49 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.50 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.51 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.52 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.53 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: ACCEPT
- expected verdict: ACCEPT
- failure total:
- witness:

11.2.54 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.55 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-TRUE

- case type: theorem_case
- observed verdict: REJECT
- expected verdict: REJECT
- failure total:
- witness:

11.2.56 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.57 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.58 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.59 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.60 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.61 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.62 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.63 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.64 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.65 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.66 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.67 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.68 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.69 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.70 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.71 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.72 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.73 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.74 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.75 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.76 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.77 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.78 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.79 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.80 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.81 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.82 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

11.2.83 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.84 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.85 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.86 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.87 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

11.2.88 FM-MIN-carrier

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.89 FM-MIN-realization

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.90 FM-MIN-lawful__possibility

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.91 FM-MIN-liveness

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.92 FM-MIN-residue

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.93 FM-MIN-morphisms

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.94 FM-MIN-boundaries

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.95 FM-MIN-operators

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.96 FM-MIN-cycles

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.97 FM-MIN-dimension

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.98 FM-MIN-k

- **case type:** component_keep_drop_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.99 FM-KLEVEL-K0_to_K1

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.100 FM-KLEVEL-K0_to_K1-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.101 FM-KLEVEL-K1_to_K2

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.102 FM-KLEVEL-K1_to_K2-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.103 FM-KLEVEL-K2_to_K3

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.104 FM-KLEVEL-K2_to_K3-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.105 FM-KLEVEL-K3_to_K4

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.106 FM-KLEVEL-K3_to_K4-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.107 FM-KLEVEL-K4_to_K5

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.108 FM-KLEVEL-K4_to_K5-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.109 FM-KLEVEL-K5_to_K6

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.110 FM-KLEVEL-K5_to_K6-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.111 FM-KLEVEL-K6_to_K7

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.112 FM-KLEVEL-K6_to_K7-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.113 FM-KLEVEL-K7_to_K8

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.114 FM-KLEVEL-K7_to_K8-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.115 FM-KLEVEL-K8_to_K9

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.116 FM-KLEVEL-K8_to_K9-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.117 FM-KLEVEL-K9_to_K10

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.118 FM-KLEVEL-K9_to_K10-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.119 FM-KLEVEL-K10_to_K11

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.2.120 FM-KLEVEL-K10_to_K11-NEG

- **case type:** adjacent_k_transition_witness
- **observed verdict:**
- **expected verdict:**
- **failure total:**
- **witness:**

11.3 Target-Blind Numeric Evidence

The empirical section reports bounded target-blind reconstruction rows over pinned official snapshots. These rows support model-core external review; they do not claim complete domain validation. - **lane total:** 5 - **failure total:** 0 - **support policy:** Rows are target-blind held-out reconstructions over pinned official snapshots. They support bounded numeric reconstruction claims only, not broad domain validation or novelty. - **domain validation policy:** NO_EMPIRICAL_PASS_FROM_OFFICIAL_SNAPSHOT_REPLAY_QA

11.3.1 Physics - OC133-TARGETBLIND-PHYSICS-001

- **snapshot:** validation/_raw/physics_nist_constants.txt
- **target-blind split:** Planck constant and speed of light rows are visible; inverse-meter joule relationship row is withheld until scoring
- **formula:** Planck_constant * speed_of_light
- **predicted value:** 1.9864458571489286e-25
- **observed value:** 1.986445857e-25
- **uncertainty:** 1e-33
- **residual:** 1.4892870576445277e-35
- **comparator baseline:** unit-incompatible Planck-constant-only negative control
- **comparator residual:** 1.9864458503739297e-25
- **negative control:** drop the speed-of-light factor and require a larger residual
- **falsifier:** Residual exceeds display-truncation tolerance or Planck-only control is not worse
- **snapshot hash:** 77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67
- **replay hash:** a83ea273ef65d6136ad7c0dfe2f7d870822fe15a062a47ae906a7d46705a0cf1
- **support scope:** target-blind reconstruction of a held-out CODATA relationship from exact defining constants; not a novel physics law

11.3.2 Chemistry - OC133-TARGETBLIND-CHEMISTRY-001

- **snapshot:** validation/_raw/chemistry_pubchem_water.txt
- **target-blind split:** formula field is visible; MolecularWeight target is withheld until scoring
- **formula:** 2*atomic_weight(H)+atomic_weight(O)
- **predicted value:** 18.01528
- **observed value:** 18.015
- **uncertainty:** 0.02
- **residual:** 0.0002800000000000058
- **comparator baseline:** CO2 molecular-weight negative control against water target
- **comparator residual:** 25.9945000000000002
- **negative control:** replace H2O by CO2 and require larger residual
- **falsifier:** Residual exceeds declared uncertainty or CO2 control is not worse than formula reconstruction
- **snapshot hash:** 316ed3babcf1b5fd6580fc2922881bf553b5971de33b72f268d943529bce293f

- **replay hash:** c115a6d84fdad856976f60ad6d2fb4c9f9c397f994c7eafb09526f944004469d
- **support scope:** target-blind reconstruction of a held-out official snapshot field; not a novel chemistry law

11.3.3 Biology - OC133-TARGETBLIND-BIOLOGY-001

- **snapshot:** validation/_raw/biology_ncbi_geo_platform.txt
- **target-blind split:** NCBI ESearch retstart/idlist fields are visible; retmax pagination target is withheld until scoring
- **formula:** retstart + len(idlist)
- **predicted value:** 20.0
- **observed value:** 20.0
- **uncertainty:** 0.0
- **residual:** 0.0
- **comparator baseline:** use total hit count as pagination-size negative control
- **comparator residual:** 43988.0
- **negative control:** replace page-size reconstruction by total hit count and require a larger residual
- **falsifier:** Retmax differs from retstart plus returned id count or total-count control is not worse
- **snapshot hash:** f205645ec097af8972f7c70f4753a5d7eb154c4283e9caa454235843974bfda4
- **replay hash:** 6606a563b35116d6852bc936618757a29873c52e59541a0928546bc863cd4ecb
- **support scope:** target-blind reconstruction of a held-out NCBI/GEO API snapshot field; not a biological mechanism law

11.3.4 Systems - OC133-TARGETBLIND-SYSTEMS-001

- **snapshot:** validation/_raw/systems_world_bank_gdp.txt
- **target-blind split:** 2021-2023 train rows predict withheld 2024 World Bank WDI target
- **formula:** GDP_2023 + (GDP_2023-GDP_2021)/2
- **predicted value:** 111042234099108.1
- **observed value:** 110982661180013.0
- **uncertainty:** 1109826611800.1301
- **residual:** 59572919095.09375
- **comparator baseline:** last-observation carry-forward GDP_2023
- **comparator residual:** 4241013358949.0
- **negative control:** last-observation baseline must have larger residual
- **falsifier:** Held-out residual exceeds 1 percent of observed target or comparator is not worse
- **snapshot hash:** 52a3dcef732b262251662f20923d00ba8d4df239cce96097abf67ee6b15ff12b
- **replay hash:** 778ffef9592179e6062fad4b770d02612408603c70455f734c423475ff704be4
- **support scope:** retrospective target-blind holdout over pinned WDI rows; not a prospective macroeconomic law

11.3.5 Mathematics - OC133-TARGETBLIND-MATHEMATICS-001

- **snapshot:** proofs/FINITE_MODEL_CHECKS_1_3_3.json
- **target-blind split:** finite theorem-case rows are visible; aggregate machine_checked_subset_total is withheld until scoring
- **formula:** count_unique(theorem_id where case_type='theorem_case' and observed_verdict='ACCEPT' and passed=true)
- **predicted value:** 10.0
- **observed value:** 10.0
- **uncertainty:** 0.0
- **residual:** 0.0
- **comparator baseline:** positive_case_total negative control, which counts non-theorem support rows too
- **comparator residual:** 11.0
- **negative control:** replace theorem-id aggregate by positive_case_total and require a larger residual

- **falsifier:** Unique accepted theorem-case count differs from machine_checked_subset_total or broad positive-case control is not worse
- **snapshot hash:** 8253425cf27d31aeb9056248447b8858370dbdaab69be6f0aed8a25b0b93f8d9
- **replay hash:** 3314f81a07054822447c104212d39cc78b79ea5b5b5c002e9ea158cb0d0131b3
- **support scope:** target-blind reconstruction of a finite proof-corpus aggregate; not a full-science program truth proof or empirical law

11.3.6 Replay Audit - Biology

- **verdict:** NUMERIC_REPLAY_QA_NOT_DOMAIN_VALIDATION
- **failure total:** 0
- **source log:** validation/numeric_predictions/OC133_NUMERIC_REPLAY_LOG.json
- **row:** OC133-NUM-BIO-GEO-COUNT
- **snapshot opened:** True
- **computed residual:** 0.0
- **negative control rejected:** True

11.3.7 Replay Audit - Chemistry

- **verdict:** NUMERIC_REPLAY_QA_NOT_DOMAIN_VALIDATION
- **failure total:** 0
- **source log:** validation/numeric_predictions/OC133_NUMERIC_REPLAY_LOG.json
- **row:** OC133-NUM-CHEM-WEBBOOK-H2O
- **snapshot opened:** True
- **computed residual:** 0.0
- **negative control rejected:** True
- **row:** OC133-NUM-CHEM-H2O
- **snapshot opened:** True
- **computed residual:** 0.0
- **negative control rejected:** True

11.3.8 Replay Audit - Mathematics

- **verdict:** NUMERIC_REPLAY_QA_NOT_DOMAIN_VALIDATION
- **failure total:** 0
- **source log:** validation/numeric_predictions/OC133_NUMERIC_REPLAY_LOG.json
- **row:** OC133-NUM-MATH-FINITE
- **snapshot opened:** True
- **computed residual:** 0.0
- **negative control rejected:** True

11.3.9 Replay Audit - Physics

- **verdict:** NUMERIC_REPLAY_QA_NOT_DOMAIN_VALIDATION
- **failure total:** 0
- **source log:** validation/numeric_predictions/OC133_NUMERIC_REPLAY_LOG.json
- **row:** OC133-NUM-PHYS-C
- **snapshot opened:** True
- **computed residual:** 0.0
- **negative control rejected:** True

11.3.10 Replay Audit - Systems

- **verdict:** NUMERIC_REPLAY_QA_NOT_DOMAIN_VALIDATION
- **failure total:** 0
- **source log:** validation/numeric_predictions/OC133_NUMERIC_REPLAY_LOG.json
- **row:** OC133-NUM-SYS-WDI-GDP
- **snapshot opened:** True
- **computed residual:** 0.0
- **negative control rejected:** True

11.4 Prior-Art Comparator and Novelty Boundary

Comparator rows are positioning evidence, not a uniqueness proof for all possible theories. Each row states overlap, residual delta, and release-safe novelty boundary. - **register status:** - **row total:** 14

11.4.1 General System Theory

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘cross-domain vocabulary’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘typed proof/data/falsifier/owner-review release governance bundle’, ‘prior_art_overlap’: ‘NOT_OBSERVED_IN_ILLUSTRATIVE_SOURCE_NOT_ABSENCE_EVIDENCE’, ‘positioning_note’: ‘auditable release-governed scientific control plane’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Ludwig von Bertalanffy, General System Theory’, ‘url’: ‘https://www.georgebraziller.com/general-systems-theory’, ‘source_date’: ‘1968’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-01-01.txt’, ‘local_protocol_snapshot_sha256’: ‘ac73ceab8486fb18e77fe2fd155ebf20’, ‘search_query’: ‘“General System Theory” “organized wholes and cross-domain system language” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTED’}]

11.4.2 Autopoiesis

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘self-producing living organization’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘residue/rebirth/identity morphism separation with owner-review claim ledger’, ‘prior_art_overlap’: ‘NOT_OBSERVED_IN_ILLUSTRATIVE_SOURCE_NOT_ABSENCE_EVIDENCE’, ‘positioning_note’: ‘typed restart/identity equivocation blocker’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY

- **source refs:** [{‘title’: ‘Maturana and Varela, Autopoiesis and Cognition’, ‘url’: ‘https://link.springer.com/book/10.1007/94-009-8947-4’, ‘source_date’: ‘1980’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-02-01.txt’, ‘local_protocol_snapshot_sha256’: ‘411971bd1277742eebee204a43f99c’, ‘search_query’: ‘“Autopoiesis” “self-production and living organization” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.3 Dynamical Systems

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘state spaces, flows, iteration’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘smooth dynamics as one typed update specialization among proof/rewrite/hybrid updates’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘anti-universal-ODE typing rule’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Encyclopedia of Mathematics, Dynamical system’, ‘url’: ‘https://encyclopediaofmath.org/wiki/Dy’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/03-01.txt’, ‘local_protocol_snapshot_sha256’: ‘3643c2233805d87a98f3b1148de0d461354dd8f06138f4a12c4476a3a9210665’, ‘search_query’: ‘“Dynamical Systems” “state evolution and flows” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.4 Category and Topos Formalisms

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘typed objects, morphisms, internal logic’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘public-release theorem/evidence/falsifier lock over typed claims’, ‘prior_art_overlap’: ‘NOT_OBSERVED_IN_ILLUSTRATIVE_SOURCE_NOT_ABSENCE_EV’, ‘positioning_note’: ‘release-machine governance over scientific claim promotion’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘nLab, topos’, ‘url’: ‘https://ncatlab.org/nlab/show/topos’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-04-01.txt’, ‘local_protocol_snapshot_sha256’: ‘14f4b28a97898ae48909f877e08070d560998f850230daf171605c47e4d1f669’, ‘search_query’: ‘“Category and Topos Formalisms” “typed objects, morphisms, categorical semantics” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_P’}]

11.4.5 RAF Theory

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘autocatalytic closure and boundary relevance’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘K3 closure as adjacent K-level with demotion and release-proof witness rows’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘classifier-level irreducibility/demotion rule’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Hordijk and Steel, Autocatalytic sets and boundaries’, ‘url’: ‘https://link.springer.com/article/10.1014-0006-2’, ‘source_date’: ‘2015’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-05-01.txt’, ‘local_protocol_snapshot_sha256’: ‘df9ee2dff8bcf86d318fb695a2e5a691’, ‘search_query’: ‘“RAF Theory” “autocatalytic closure and boundaries” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.6 Complexity and Information Measures

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.

- **feature tests:** [{‘oc_feature’: ‘information/complexity quantities’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘historical-axis versus effective-rank distinction inside OC K-level claims’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘typed anti-conflation theorem and finite witness’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Stanford Encyclopedia of Philosophy, Information’, ‘url’: ‘https://plato.stanford.edu/entries/infor’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/06-01.txt’, ‘local_protocol_snapshot_sha256’: ‘004e1d650f029f27ba8cd0cf99cab8fc07cb50e83bd7ab423e836b398544f25’, ‘search_query’: ‘“Complexity and Information Measures” “information-theoretic and complexity quantities” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.7 Causal and Identity Theories

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘persistence and identity criteria’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘residue/rebirth never promoted as identity without explicit invariant preservation’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘release claim-boundary lock’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Stanford Encyclopedia of Philosophy, Identity Over Time’, ‘url’: ‘https://plato.stanford.edu/entries/time/’, ‘source_date’: ‘2026’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/07-01.txt’, ‘local_protocol_snapshot_sha256’: ‘ffbde0b100aeb2986e68c25b2ef79e78a2a842c92d6f03f7b53969e4b6ccdef4’, ‘search_query’: ‘“Causal and Identity Theories” “diachronic identity and persistence problems” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.8 Systems Engineering

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘requirements, verification, validation, lifecycle thinking’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘owner-gated owner-review scientific release state as theorem/evidence control plane’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘scientific publication lock integrated with claim ledger’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘INCOSE, Systems Engineering and System Definitions’, ‘url’: ‘https://www.incose.org/about-systems-engineering/system-and-se-definitions/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-08-01.txt’, ‘local_protocol_snapshot_sha256’: ‘c279b39dbb5291d70c10d2493fbd7d5d2cd8cc1c1d067609b889ebd7373bf809’, ‘search_query’: ‘“Systems Engineering” “verification, validation, lifecycle governance” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.9 Hybrid Systems

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘continuous/discrete hybrid transition systems’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘hybrid operator claim used to block universal differential overreach’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘claim-boundary role in OC operator theorem’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Hybrid Systems III, Springer’, ‘url’: ‘https://link.springer.com/book/10.1007/BFb0031987’, ‘source_date’: ‘1996’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-09-01.txt’, ‘local_protocol_snapshot_sha256’: ‘53321c329f66879221e61823275257e51d3c1e008cb5104bae6f7e9e2676a8b6’, ‘search_query’: ‘“Hybrid Systems” “hybrid continuous/discrete transitions” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.10 Formal Methods and Lean

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘machine-checked proof development’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘Lean subset plus finite semantic witnesses plus release gates for public claim promotion’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘artifact-bound scientific release policy’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Lean 4 official site’, ‘url’: ‘https://lean4.dev/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-10-01.txt’, ‘local_protocol_snapshot_sha256’: ‘6d3b6cc5afbe2d8bae85a6cd7caf72f4f52d4a7195cc4d85fcd69346b8fb668b’, ‘search_query’: ‘“Formal Methods and Lean” “theorem proving and formal verification” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.11 Assurance Cases and Safety Cases

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘structured argument that evidence supports a claim’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘OC v12 claim ledger plus owner-review release gates’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘scientific-release control plane around theory claims’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘NASA System Safety Handbook, safety case/assurance case practice’, ‘url’: ‘https://www.nasa.gov/reference/system-safety-handbook/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-30’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-11-01.txt’, ‘local_protocol_snapshot_sha256’: ‘9c12be7e0cac872baf6e93547178bb857cee41b88d8c1d4e9001d548b0cba2b2’, ‘search_query’: ‘“Assurance Cases and Safety Cases” “assurance case argument/evidence structures” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.12 Goal Structuring Notation / Argument Patterns

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘claim -> argument -> evidence trace’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘machine-generated attack matrix and finite-case route map’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘local scientific release binding; no uniqueness promoted’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘The Assurance Case Working Group and GSN community references’, ‘url’: ‘https://scsc.uk/scsc-141B’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-30’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-12-01.txt’, ‘local_protocol_snapshot_sha256’: ‘ee43f5a19310487c8c6ac885bfe9068ec6cb40e3e1c303c9edc54fce5913b861’, ‘search_query’: ‘“Goal Structuring Notation / Argument Patterns” “structured claim/evidence argument notation” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.4.13 Requirements Traceability and V&V Matrices

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘requirement/claim to verification artifact traceability’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘theorem-to-finite-case-to-Cerberus-finding closure ledger’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘local owner-review scientific release workflow’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY

- **source refs:** [{‘title’: ‘NASA Systems Engineering Handbook’, ‘url’: ‘https://www.nasa.gov/reference/nasa-systems-engineering-handbook/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-30’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-13-01.txt’, ‘local_protocol_snapshot_sha256’: ‘9c24dc9b859c4447b534b035086318467e6cc072cbc5d5b711b10073a9eab584’, ‘search_query’: “Requirements Traceability and V&V Matrices” “requirements traceability and verification/validation matrices” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTED’}]

11.4.14 Scientific Workflow Provenance and Reproducibility Systems

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘artifact metadata, provenance, and reproducibility package’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘strict owner-review theorem/replay/Cerberus gate stack’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘local policy coupling; no uniqueness promoted’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘RO-Crate specification’, ‘url’: ‘https://www.researchobject.org/ro-crate/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-30’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/14-01.txt’, ‘local_protocol_snapshot_sha256’: ‘b71da73a5a5235fca9d08148a048225ad33f79da1bfc73eae920e3b9d0daff5’, ‘search_query’: “Scientific Workflow Provenance and Reproducibility Systems” “workflow provenance, reproducibility metadata, and research-object packaging” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

11.5 Phenomenon Coverage and Limits

The phenomenon matrix records model cards and evidence routes. Rows with illustrative or protocol-ready status are not promoted as complete phenomenon explanations. - **row total:** 15 - **state:**

11.5.1 P001

- **observable:** same-cell raw pair is not distinguished; cross-cell quotient pair is distinguished
- **negative control:** FM-T133-K0-RES-NEG
- **falsifier:** If a same-rho-cell raw pair is accepted as resolution-distinguished, the K0 resolution boundary fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.2 P002

- **observable:** death blocks live status; rebirth source is residue and target is a new live token
- **negative control:** FM-T133-OMEGA-STATUS-NEG
- **falsifier:** If death=true and live=true are accepted together, or rebirth targets the original identity token, the lifecycle theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.3 P003

- **observable:** live label fails without cycle or maintenance support
- **negative control:** FM-T133-CYCLE-NEG
- **falsifier:** If live=true is accepted with cycle_mode=none and maintenance support unavailable, the liveness route fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.4 P004

- **observable:** classifier failure equals declared failure predicate; metric wording is allowed only when a measure is declared
- **negative control:** FM-T133-BOUNDARY-NEG
- **falsifier:** If metric boundary language is accepted without `metric_measure_declared=true`, the boundary specialization fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.5 P005

- **observable:** proof/rewrite update is accepted only as a typed non-smooth transition
- **negative control:** FM-T133-HYBRID-PROOF-UPDATE-NEG
- **falsifier:** If a proof/rewrite state can request a derivative without a smooth chart, the operator boundary fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.6 P006

- **observable:** historical activation remains while effective rank drops
- **negative control:** FM-T133-DIM-NEG
- **falsifier:** If a historical axis can decrease when effective rank drops, the dimension theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.7 P007

- **observable:** $k=0$ is licensed by declared zero-cause, not empty state set
- **negative control:** FM-T133-K-ZERO-NEG
- **falsifier:** If $k=0$ is accepted with nonempty support and no declared zero-cause, the k -zero theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.8 P008

- **observable:** closure cannot be reduced when production witness remains observable
- **negative control:** FM-KLEVEL-K2_to_K3-NEG
- **falsifier:** If $K2 \rightarrow K3$ reduction preserves the production witness while still demoting, the closure-like K transition fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.9 P009

- **observable:** $K6 \rightarrow K7$ transition fails reduction when role witness changes verdict
- **negative control:** FM-KLEVEL-K6_to_K7-NEG
- **falsifier:** If a role/norm witness changes allowed action but the $K6 \rightarrow K7$ reduction still passes, the institution model card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.10 P010

- **observable:** $K8 \rightarrow K9$ transition fails reduction when claim revision is enabled
- **negative control:** FM-KLEVEL-K8_to_K9-NEG

- **falsifier:** If claim/evidence revision changes verdict but K8->K9 reduction still passes, the theory-change card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.11 P011

- **observable:** self-application is accepted only through typed transition witness
- **negative control:** FM-KLEVEL-K9_to_K10-NEG
- **falsifier:** If self-application is admitted without the K9->K10 typed transition witness, the recursion card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.12 P012

- **observable:** public action is rejected while global owner-review or any channel lock remains closed
- **negative control:** ADV-NOSEND-PUBLISH-HYPOTHETICAL-OWNER-APPROVED-CONTROL
- **falsifier:** If publish is allowed while owner approval is absent or any channel lock remains not authorized, the owner-review card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.13 P013

- **observable:** reduction fails exactly when retained witness stays observable
- **negative control:** FM-T133-KLEVEL-NEG
- **falsifier:** If any adjacent K retained-witness row lacks its demotion control, the K-collapse card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.14 P014

- **observable:** component removal changes declared verdict in semantic finite runner
- **negative control:** FM-T133-MIN-NEG
- **falsifier:** If a tuple component can be removed without the semantic finite verdict changing, the minimality card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.5.15 P015

- **observable:** identity continuation is accepted only when endpoint-bound identity evidence is present
- **negative control:** FM-T133-ID-NEG
- **falsifier:** If residue or rebirth evidence is accepted as identity continuation without endpoint-bound identity evidence, the identity theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

11.6 Formal Spine and Appendices

11.6.1 content/03_model.tex

% FILE: content/03_model.tex

11.6.2 Ontological Structure of the Model

This section presents the formal core of the Ontology of Continua (OC). It gives the reader the unified notation, axioms, definitions, and structural constraints needed for the later chapters on closure, synthesis, and practical use.

OC describes continua across multiple scientific domains through one structural language. That language is built from axes, potentials, flows, thresholds, boundaries, cycles, and a measure of continuumness. Each continuum is also embedded in a surrounding meta-space. All definitions in this section are domain-independent at the level of formal vocabulary. Later review chapters will sort claims by proof status and empirical support; this section supplies the common model language those checks depend on.

11.6.3 Axiomatic foundation: Level K_0

Level K_0 is a purely structural substrate. It is not a physical space; it carries no time, energy, geometry, or dynamics. Its role is to provide the minimal conditions under which any higher-level continuum is logically possible.

11.6.4 Specification of K_0

K_0 is specified by a resolution-relative tuple

$$K_0 = (S, R, \Delta_{\rho}, C),$$

where:

- S is a raw carrier of states, which may be finite, countable, continuous, or typed-combinatorial;
- R is the family of admissible resolution regimes;
- each $\rho \in R$ induces a quotient $S_{\rho} = S / \sim_{\rho}$ and a structural difference function $\Delta_{\rho}: S_{\rho} \times S_{\rho} \rightarrow \mathbb{R}_{\geq 0}$;
- C is a structural relation (or family of relations) preserving distinguishability.

There is no time parameter and no evolution operator at this level.

11.6.5 Axiom 0.1 (Difference and distinguishability).

For every resolution regime $\rho \in R$ and all resolved classes $[s_1]_{\rho}, [s_2]_{\rho} \in S_{\rho}$,

$$\Delta_{\rho}([s_1]_{\rho}, [s_2]_{\rho}) = 0 \iff [s_1]_{\rho} = [s_2]_{\rho}.$$

Nonzero resolved structural difference is the minimal condition for distinguishability at regime ρ . A continuum cannot exist without at least two states distinguishable in some declared resolution regime.

paragraph Axiom 0.2 (Resolution-relative threshold Θ_0, ρ). For each active resolution regime ρ there exists $\varepsilon_{\rho} > 0$ such that

$$\text{forall } [s_1]_{\rho}, [s_2]_{\rho} \in S_{\rho}, [s_1]_{\rho} \neq [s_2]_{\rho} \implies \Delta_{\rho}([s_1]_{\rho}, [s_2]_{\rho}) \geq \varepsilon_{\rho}$$

The value $\Theta_0, \rho = \varepsilon_{\rho}$ is a threshold on resolved classes, not on raw states. Raw points inside a single ρ -class may be arbitrarily close or even continuous; no theorem in Core-1.3.3 may infer global raw-state uniform discreteness from K_0 .

11.6.6 Axiom 0.3 (Logical substrate).

K_0 carries no time parameter and no dynamical operator. It does not evolve and does not generate higher levels by itself. It specifies only logical conditions on distinguishability and the existence of nontrivial differences. All dynamics and all notions of energy, geometry, and causality belong to levels K_1 and above.

In particular, no operator defined on K_0 can increase dimension: differences at level K_0 are always expressed along the existing structural axis of distinguishability. The dimensionality associated with K_0 is fixed by the embedding meta-space M_0 . Here M_0 denotes the minimal meta-space in which the purely structural

substrate can be evaluated for later K_0 to K_1 emergence; it is not a physical container or a dynamical environment.

11.6.7 Construction of Level K_1

Level K_1 is the simplest genuine continuum: it introduces time, a one-dimensional axis, and basic geometric structure.

11.6.8 Specification of K_1

The continuum K_1 is defined by:

$$K_1 = \text{big}(X_1, \tau_1, \Omega_1, A_1, P_1(t), J_1(t), \Theta_1, \text{partial}\Omega_1, C_1, k_1(t) \text{big}),$$

where:

- A_1 is a single axis (a one-dimensional coordinate);
- (X_1, τ_1) is a topological space obtained from the structural data of K_0 , typically an interval $X_1 = (a, b)$;
- Ω_1 is a space of admissible configurations on X_1 with appropriate regularity conditions. In this time-dependent analytic example,

$$\Omega_1 = C^{0(I, H^1(X_1, V_1))} \cap C^{1(I, L^2(X_1, V_1))},$$

where I is the time interval and V_1 is the Hilbert space used by the example; a finite-dimensional vector space is the special case used when the local model has finitely many degrees of freedom. - $P_1(t)$ is an energy-like potential functional on Ω_1 ; - $J_1(t)$ are flows derived from the variation of $P_1(t)$; - Θ_1 encodes minimal conditions for classical stability; - $\text{partial}\Omega_1$ is the boundary where stability thresholds saturate; - C_1 is the set of classical cycles (for example, periodic orbits); - $k_1(t)$ measures one-dimensional continuumness according to the general definition below.

paragraphTransition 0to 1Psi_0to_1 The passage K_0 to K_1 is generated by an operator

$$\text{Psi}_0 \text{to } 1 : (S, \Delta, C) \text{ mapsto } (X_1, \tau_1, A_1, \Omega_1, \Theta_1),$$

which constructs the skeletal K_1 scaffold; the energy functional, flow family, boundary, cycles, and continuumness scalar are specified by the subsequent K_1 -specific clauses rather than by this first transition map alone. The scaffold construction:

- identifies a family of structurally compatible states in S ;
- equips them with an order and topology (X_1, τ_1) ;
- defines the first axis A_1 ;
- constructs the admissible configuration space Ω_1 ;
- induces classical thresholds Θ_1 .

This is the first instance of dimensional emergence: a continuous axis appears that cannot be represented within the purely structural substrate of K_0 .

11.6.9 General definition of a continuum

For any level K in the hierarchy, a continuum is defined as the tuple

$$K = \text{big}(\Omega(K), A(K), P(t), J(t), \Theta(K), \text{partial}\Omega(K), C(K), k(K, t) \text{big}),$$

where:

- $\Omega(K)$ is a nonempty set of admissible states;
- $A(K) = \set{A_1, \dots, A_n}$ is a finite set of axes of incompatible differences;
- $P(t)$ is the vector of potentials on $\Omega(K)$;
- $J(t)$ is the aggregate potential-flow vector; state updates are handled by the evolution operator E ;
- $\Theta(K) = \set{\Theta_i}$ is the set of threshold functions;
- $\text{partial}\Omega(K)$ is the boundary of admissible states;
- $C(K)$ is the family of structurally stable cycles;

- $k(K,t)$ is the measure of continuumness.

Every continuum is assumed to be embedded in a surrounding meta-space M such that

$$\Omega(K) \subseteq \Omega(M), \quad A(K) \subseteq A(M).$$

The meta-space provides additional admissible states and axes that can host future dimensional extensions of K .

11.6.10 State space and boundary

Boundaries are represented first as typed admissibility classifiers

$$b_i: \Omega_{\text{cand}}(K) \rightarrow \text{Status}_i, \quad \text{fail}_i: \text{Status}_i \rightarrow \{\text{top}, \text{bot}\}.$$

The admissible state set is

$$\Omega(K) = \{s \in \Omega_{\text{cand}}(K) \mid \neg \text{fail}_i(b_i(s)) \text{ for all } i\}.$$

The older real-valued threshold notation is the quantitative special case in which $\text{Status}_i = \mathbb{R}$ and $\text{fail}_i(x)$ means $x > 0$. In that special case one may write functions $f_i: \Omega(K) \rightarrow \mathbb{R}$ with

$$f_i(s) \leq 0 \quad \text{for all } s \in \Omega(K),$$

and the boundary is defined as

$$\partial\Omega(K) = \{s \in \Omega(K) \mid \exists i, f_i(s) = 0\}.$$

This displayed boundary equation is therefore not primitive for all domains. It is the metric or real-valued specialization of the classifier boundary and must not be used to force logical, proof-theoretic, social, or categorical boundaries into artificial numerical geometry.

The boundary is dynamic. The notation below is an update map, not a derivative of a set-valued object:

$$\partial\Omega(K, t+dt) = \text{Rbig}(\partial\Omega(K, t), P(t), J(t), \Theta(K)),$$

which can contract, expand, or bifurcate $\Omega(K)$ during birth, life, and death events.

11.6.11 Taxonomy of thresholds

Each continuum has a structured set of thresholds $\Theta(K)$, organised into the following types:

- Existence thresholds Θ_{exist} : conditions under which $\Omega(K) \neq \emptyset$.
- Stability thresholds Θ_{stab} : conditions for bounded dynamics and non-divergent flows.
- Critical thresholds Θ_{crit} : hypersurfaces where qualitative changes (phase transitions, bifurcations) occur.
- Dimensional thresholds Θ_{dim} : conditions under which new axes and higher-dimensional continua can emerge.
- Death thresholds Θ_{death} : structural limits where no admissible states remain and $\Omega(K)$ collapses.

The full threshold landscape of a continuum is thus a collection of inequalities:

$$f_i^{\text{exist}}(s) \leq 0, \quad f_i^{\text{stab}}(s) \leq 0, \quad f_i^{\text{crit}}(s) \leq 0, \quad f_i^{\text{dim}}(s) \leq 0, \quad f_i^{\text{death}}(s) \leq 0,$$

with corresponding boundary components given by the equalities.

11.6.12 Potentials, flows, and structural tension

Potentials $P(t)$ encode the internal configuration of constraints and driving forces within a continuum. They may correspond to energy landscapes, chemical concentrations, membrane gradients, representational or informational structures, or institutional and normative pressures.

The aggregate flow vector $J(t)$ describes typed potential update. In smooth realizations this update can be written as a derivative,

$$dPdt = J(t),$$

but in discrete, proof-replay, stochastic, logical, or governance realizations J is a transition/update component rather than a differential object. With that typing fixed, $J(t)$ decomposes into structurally distinct components:

- supporting flows $J_support$, which maintain cycles and stabilise $k(K,t)$;
- critical flows $J_critical$, which move the system towards or across critical thresholds Θ_crit and Θ_dim ;
- destructive flows J_kill , which push the system across Θ_death and reduce $k(K,t)$.

Structural tension $T(K,t)$ is a functional of potentials, axes, and gradients (schematically $T=T(P,A,nabla P)$). It measures how strongly the current configuration stresses the threshold landscape. Dimensional transitions occur when $T(K,t)$ exceeds $\Theta_dim(K)$; collapse occurs when destructive flows combined with tension drive the system across $\Theta_death(K)$.

11.6.13 Cycles and continuumness

Cycles $C(K)$ are declared cycle modes: maintenance recurrences, renewal loops, proof replays, regulatory updates, periodic orbits, or degenerate fixed-point maintenance checks. A closed trajectory in $\Omega(K)$ is only the smooth or geometric special case. When a quantitative trajectory realization is declared, cycles may satisfy

$$L(C) = \int_C \delta L_K < \infty, \text{quad } S(C) = \min_s \int_C \delta \partial \Omega(s) > 0,$$

where δL_K is the chosen structural length element for the realised continuum and $\delta \partial \Omega(s)$ is the corresponding boundary clearance. The classifier boundary definition above is metric-free; these cycle estimates add a structural gauge only when quantitative clearance or length is being measured. In metric realisations this gauge may be an induced distance. In non-metric realisations it is any level-admissible separation functional that vanishes exactly on the boundary and is positive on structurally interior states.

When the continuum evolves, the admissible region is read as time-dependent: $\Omega(K,t)$ denotes the state space available to K at time t , while $\Omega(K)$ denotes the underlying level-relative admissible space when no time slice is being specified. When older formulas write $\Omega(K(t))$, read that notation as the same time-slice object $\Omega(K,t)$. The model adopts the following unified definition of continuumness:

$$k(K,t) = \chi_Omega(s_K(t);K,t); S_axes(K,t); S_cycles(K,t); S_flows(K,t); S_coh(K,t),$$

where:

- $s_K(t)$ is the current state of K at time t , and $\chi_Omega(s_K(t);K,t)$ is the pointwise admissibility indicator: it is 1 if $s_K(t) \in \Omega(K,t)$, and 0 otherwise;
- $S_axes(K,t)$ quantifies effective axis saturation, e.g. the ratio of the effective rank of working axes to the maximal possible rank for that level;
- $S_cycles(K,t)$ measures the strength and efficiency of stable cycles, such as a weighted sum of cycle efficiencies normalised by their maximal values;
- $S_flows(K,t)$ captures flow stability. With aggregated supporting and destructive magnitudes $\Phi_support$ and Φ_kill , one local rule is shown below. The zero-flow branch applies only when the tested state has no active flow channel:

$$S_flows(K,t) =$$

$$\frac{\Phi_support(t)}{\Phi_support(t) + \Phi_kill(t)}, \& \Phi_support(t) + \Phi_kill(t) > 0, \& \Phi_support(t) = \Phi_kill(t)$$

Thus if support vanishes while destructive flow is positive, $S_flows=0$; $-S_coh(K,t) \in [0,1]$ encodes global structural coherence, which may include connectedness of the relevant graphs, compatibility of thresholds and flows, and absence of contradictory constraints.

All multiplicative factors in the displayed definition are normalised to $[0,1]$ in the local model before the product is evaluated. The continuumness-update operator U , developed later in Section~sec:operators-full, advances $k(K,t)$ to the next lawful time slice using these factors.

By construction $0 \leq k(K,t) \leq 1$. Positive $k(K,t)$ is the live-state indicator. It records four simultaneous facts: the current state is admissible, the active axes remain supported, the cycle factor and flow-stability factor

both remain positive, and the coherence factor has not collapsed. The full live-state status also requires the admissible-state, threshold, and identity clauses stated in Section~sec:collapse-rebirth. Death corresponds to $k(K,t)$ to 0 in combination with the collapse of $\Omega(K)$.

11.6.14 Evolution operator

The evolution of a continuum is described at the structural level by an operator

E: $K(t)$ mapsto $K(t+dt)$,

or, in expanded form,

E: $(\Omega, A, P, J, \Theta, \text{partial}\Omega, C, k) \text{ longrightarrow } (\Omega', A', P', J', \Theta', \text{partial}\Omega', C', k')$.

Here A' denotes the post-evolution axis set of the same continuum in the expanded tuple. This is a unary evolution operator on a time slice of K . The active meta-space belongs to the surrounding environment. When it must be named locally, it is shown as a subscripted parameter rather than as a second argument of E, so the operator arity remains fixed.

The model does not merge flow updates, threshold tests, cycle structure, boundary motion, axis extension, and continuumness updates into one differential system. It treats these components as coupled parts of a constrained structural evolution framework with the following requirements:

- Consistency: if $s(t)$ in $\Omega(K(t))$, then either $s(t+dt)$ in $\Omega(K(t+dt))$ or the transition is associated with a threshold crossing.
- Threshold-respecting: flows obey the inequality structure encoded in $\Theta(K)$, except when explicitly crossing critical or death thresholds.
- Monotonicity of dimension: if A' contains a new axis not representable as a combination of previous axes, then $\dim(K(t+dt)) > \dim(K(t))$; dimension cannot decrease.

The dynamics continue as long as $\Omega(K(t)) \neq \text{emptyset}$. Once $\Omega(K(t^{\wedge})) = \text{emptyset}$, the continuum is dead and E can no longer act meaningfully on it.

11.6.15 Embedding into meta-spaces

Each level K_x is embedded in a meta-space M_x that provides additional admissible states and axes. At the structural level this is captured by the conditions

$\Omega(K_x(t)) \text{ subseteq } \Omega(M_x(t)), \text{ qquad } A(K_x) \text{ subseteq } A(M_x)$.

The universal evolution axiom can then be stated schematically as

$K_x(t+dt) = E_{M_x}(t)(K_x(t))$,

with the additional requirement that if the time-slice embedding condition fails, $\Omega(K_x(t)) \not\text{subseteq } \Omega(M_x(t))$, the continuum K_x ceases to exist. When the time slice is clear, the shorter notations K_x and M_x suppress the argument t . Dimensional growth of K_x is only possible if the meta-space contains an axis A_{new} in $A(M_x) \setminus A(K_x)$ and structural tension exceeds the corresponding dimensional threshold. This is the axis-origin constraint used below: new axes cannot be generated from K_x alone.

11.6.16 Birth of continua

The emergence of a new continuum K_{x+1} from K_x is a threshold-induced phase transition. Structurally, birth occurs when the following conditions are met:

- A new class of differences appears that cannot be represented using the existing axes $A(K_x)$.
- Structural tension associated with these differences satisfies

$T(K_x, t) > \Theta_{\dim}(K_x)$.

- The embedding space M_x contains at least one axis that can host the new differences (i.e. A_{new} in $A(M_x) \setminus A(K_x)$).

- There exists a nonempty set of admissible states $\Omega(K_{x+1})$ compatible with the new axis and thresholds.

The operator of dimensional birth

$\Psi_{x \rightarrow x+1}: K_x \rightarrow K_{x+1}$

is minimal and irreversible: any nonzero emergence of the new axis constitutes the new continuum K_{x+1} , and the dimension of K_{x+1} cannot revert to that of K_x without destroying $\Omega(K_{x+1})$. The local model therefore instantiates dimensional monotonicity. The corresponding no-spontaneous-creation result is recorded separately in Section~sec:theorem-spontaneous-dimension-creation.

11.6.17 Life of continua

A continuum K is alive on an interval of time if

$\Omega(K(t)) \neq \emptyset, \forall t \in I, \text{ and } C(K(t)) \neq \emptyset,$

and if supporting flows J_{support} dominate destructive flows J_{kill} when integrated over relevant cycles.

Life thus corresponds to the persistent existence of:

- stable cycles separated from the boundary by a finite structural distance;
- sufficient supporting flows to maintain these cycles;
- coherence between potentials, axes, and thresholds;
- controlled critical behaviour that does not cross Θ_{death} .

These conditions are interpreted differently at each level K_x , but the structural pattern is the same from protocells to institutions.

11.6.18 Death of continua

A continuum K dies at time t^{ast} when

$\Omega(K(t^{\text{ast}})) = \emptyset,$

which implies $\chi_{\Omega}(s_{K(t^{\text{ast}})}; K, t^{\text{ast}}) = 0$ and hence $k(K, t^{\text{ast}}) = 0$. Equivalently:

- all stable cycles vanish, $C(K(t^{\text{ast}})) = \emptyset;$
- no state satisfies the threshold inequalities concurrently;
- any attempted continuation of dynamics would violate at least one existence threshold Θ_{exist} .

11.6.19 Irreversibility of death

Once $\Omega(K(t^{\text{ast}})) = \emptyset$, there is no structural operator acting within the same level that can reconstruct a nonempty $\Omega(K)$. Any apparent resurrec

11.6.20 content/OC_1_3_3_TYPED_FOUNDATION.tex

11.6.21 OC 1.3.3 Typed Foundation

The v12 foundation treats an OC realization as a typed tuple

$R = (S, \rho, \Omega, L, D, E, M, B, O, C, A, k)$

where S is the raw carrier, ρ is the resolution regime, Ω is lawful possibility, L is time-sliced liveness, D is death status, E is residue, M is the morphism family, B is generalized boundary data, O is typed operator semantics, C is cycle mode, A is the historical/effective dimension record, and k is the continuumness functional with explicit zero-cause records.

No public theorem is allowed to use a symbol before its type, carrier, realization, and failure mode have been declared. Raw continuity, quotient distinguishability, liveness, residue, rebirth, and identity are not synonyms. A realization that erases one of these distinctions fails the v12 claim-boundary gate.

11.6.22 `content/OC_1_3_3_OPERATOR_SEMANTICS.tex`

11.6.23 OC 1.3.3 Operator Semantics

Operators F,G,H,Q,R,S,U are typed update components. In v12, a chart-labelled route is only flow-one notation over a declared chart/domain/local-law record; it is not an ODE solution theorem, not a vector-field theorem, and not a differentiability theorem. Proof/rewrite and guard/reset routes are first-class non-smooth update routes. Stochastic, graph, and institutional operators are listed only as extension obligations until separate formal and finite evidence exists.

The v12 package rejects those stronger readings at the claim-boundary level. A future release may add a corpus-wide lint gate for differentiability/manifold/vector-field/ODE wording, but v12 does not claim such a lint theorem; those stronger mathematical claims require route-specific evidence first.

11.6.24 `content/OC_1_3_3_CYCLE_TAXONOMY.tex`

% FILE: content/OC_1_3_3_CYCLE_TAXONOMY.tex ### OC 1.3.3 Cycle Taxonomy and Liveness Modes

Core 1.3.3 distinguishes dynamical cycles, recurrence cycles, maintenance cycles, fixed-point or degenerate cycles, constraint-satisfaction loops, institutional feedback loops, and proof/program replay loops. A stable fixed point is not excluded from liveness when it supplies a degenerate maintenance cycle under the level's identity predicate. Dormant and frozen states are live or latent when carrier identity and maintenance constraints persist; they are dead only when no live admissible realization remains.

11.6.25 Theorem.

For each K-level, liveness requires the cycle mode specified by its carrier and identity condition, not necessarily a smooth closed trajectory.

11.6.26 Proof.

The K-level matrix declares the required cycle mode. If the mode is satisfied, the liveness predicate has recurrent or maintenance support. If it is absent, the carrier lacks the refresh, feedback, replay, or constraint loop needed to preserve identity. Thus stable equilibria, crystals, archived theories, and dormant organisms are classified by their maintenance or residue status rather than by a narrow periodic-orbit test.

11.6.27 `appendix/OC_1_3_3_K0_RESOLUTION_FOUNDATION.tex`

11.6.28 K0 Resolution Foundation

K0 distinguishability is defined over S/ρ , not over raw S . The quotient may be finite even when S is continuous. Uniform separation is therefore a property of resolved cells, not a hidden atomistic ontology.

Finite witness: partition $[0,1]$ into four observation cells. Raw points inside one cell are not distinguished; two cells are distinguished by the quotient index.

11.6.29 `appendix/OC_1_3_3_CONTINUUMNESS_FUNCTIONALS.tex`

11.6.30 Continuumness Functionals

The primitive statement is not the old biconditional $k=0$ iff Ω or cycles are empty. The v12 primitive is $k=0$ iff at least one declared zero-cause predicate is active. Zero causes include admissibility failure, flow collapse, coherence contradiction, identity break, and embedding failure. Product formulas are local aggregators after these causes are typed.

11.6.31 `appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex`

11.6.32 Boundary Representation Theorem

A boundary is a family of classifiers $b_i: \text{Sto } T_i$ plus failure predicates on T_i . Metric thresholds are the special case where T_i is ordered numeric data. Logical, categorical, graph, proof-state, and institutional boundaries

do not need fake real-valued surfaces.

11.6.33 appendix/OC_1_3_3_VERDICT_INVARIANT_MINIMALITY_FULL_PROOF.tex

% FILE: appendix/OC_1_3_3_VERDICT_INVARIANT_MINIMALITY_FULL_PROOF.tex % OC Core 1.3.3 scientific closure artifact.

11.6.34 OC 1.3.3 Verdict-Invariant Minimality Full Proof

11.6.35 Claim.

For the OC verdict class used in Core 1.3.3, removing any required component from the tuple changes at least one admissibility, liveness, collapse, identity, or release-governance verdict.

11.6.36 Witness method.

Each component has a witness pair (K,K') that agrees on all other components and differs on one verdict only because the removed component is no longer observable.

11.6.37 Proof.

The witness matrix in `proofs/minimality/WITNESS_PAIRS.json` supplies pairs for $\Omega, A, P, J, \Theta, \text{partial}\Omega, C, k$, carrier, residue, and morphism support. For each row the two structures are isomorphic after forgetting the target component and non-isomorphic with respect to the target verdict before forgetting it. Therefore the component is verdict-relevant. Since every required component has such a witness, the tuple is minimal for this verdict class.

11.6.38 appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex

11.6.39 K-Level Irreducibility Atlas

The v12 atlas is adjacent and witness-based. A K-level is irreducible exactly when its adjacent witness remains observable under the declared release equivalence. llll Transition & Added axis & Witness & Demotion criterion \ K0_to_K1 & distinguishable state -> minimal continuum & continuity obligation changes verdict & demote if no continuity obligation is observed \ K1_to_K2 & minimal continuum -> phase threshold & threshold crossing changes admissibility & demote if threshold never affects verdict \ K2_to_K3 & phase threshold -> autocatalytic closure & closure production is required & demote if closure production is irrelevant \ K3_to_K4 & closure -> membrane boundary & inside/outside classifier changes verdict & demote if boundary classifier has no observable effect \ K4_to_K5 & boundary -> excitable regulation & signal-triggered update changes verdict & demote if excitation never affects status \ K5_to_K6 & regulation -> binding prediction & binding relation changes next-state prediction & demote if binding is observationally inert \ K6_to_K7 & binding -> trust coordination & norm/role classifier changes allowed action & demote if roles do not change allowed actions \ K7_to_K8 & coordination -> regime shift & meta-state transition changes future rules & demote if regime state is constant \ K8_to_K9 & regime -> theory dynamics & claim/evidence update changes theory verdict & demote if claim revision is disabled \ K9_to_K10 & theory dynamics -> recursive self-application & model applies to its own updates & demote if self-reference is absent \ K10_to_K11 & recursion -> cross-domain coherence & translation invariant changes verdict & demote if no cross-domain bridge exists \ K11_to_K12 & cross-domain coherence -> self-auditing evidence governance & evidence-policy state changes theorem promotion verdict & demote if self-audit policy state is inert \

Reduction-failure criteria are stored machine-readably in `data/k_level_irreducibility_matrix.json` and executed in `proofs/FINITE_MODEL_CHECKS_1_3_3.json`. Historical axis activation and effective working rank remain separate quantities.

11.6.40 appendix/OC_1_3_3_COMPARATOR_AND_NOVELTY_MATRIX.tex

% FILE: appendix/OC_1_3_3_COMPARATOR_AND_NOVELTY_MATRIX.tex % OC Core 1.3.3 scientific closure artifact.

11.6.41 OC 1.3.3 Comparator and Novelty Matrix

11.6.42 Comparator rule.

OC is not claimed to replace general systems theory, autopoiesis, dynamical systems, category theory, complex-systems metrics, RAF theory, or institutional analysis. Its promoted novelty is the typed cross-domain packaging of admissibility, boundary, liveness, residue, rebirth, proof/replay governance, and release-surface traceability.

11.6.43 Honesty clause.

Where a neighboring theory already supplies stronger domain mathematics, OC acts as a typed integration and governance layer, not as a superior replacement.

11.7 Journal Owner-Review Packages

Eight venue packets are included for owner review. They are not submitted by this release action. Each packet contains a package manifest, cover letter draft, checklist, reproducibility/data statement, conflict/funding statement, AI assistance disclosure, and venue-fit note. - **package total: 8 - recommended package total: 2**

11.7.1 FOUNDATIONS_OF_SCIENCE

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.2 SYNTHESIS

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.3 FOUNDATIONS_OF_PHYSICS

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.4 PHYSICAL_REVIEW_RESEARCH

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.5 ACS_OMEGA

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.6 ACTA_BIOTHEORETICA

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW

- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.7 PLOS_COMPUTATIONAL_BIOLOGY

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

11.7.8 GLOBAL_JOURNAL_OF_FLEXIBLE_SYSTEMS_MANAGEMENT

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**